

Supplementary Material for Degradation-Conditioned Observation Modeling for Multi-Object Tracking under Photometric Saturation

1 Dataset annotation and quality audits

The double-annotation audit contains 48,359 target-frame annotations sampled across all annotation-provenance categories, 312 glare events, and a dedicated sample of competitive events. Agreement is measured before adjudication. Box center and boundary errors are normalized by the reference box diagonal. The competitive row in Table 1 is an overlapping subset of the visibility categories. Presence-only intervals have no spatial box and therefore contribute only presence, identity, and adjudication measures.

Table 2 reports the remaining dataset-quality checks. Reference-view registration is evaluated on held-out correspondences not used to fit the registration transform. Event-proposal recall is estimated with stratification weights from complete clips rejected by the automatic candidate mask; the 11 missed glare events identified by the audit are added to the benchmark. Calibration-correspondence precision is measured by manual review of a stratified 2% sample of automatically accepted matches and misses.

The lower agreement under reference-supported and competitive glare motivates the exclusion of unresolved cases from spatial calibration and identity evaluation. After adjudication, all retained calibration samples have an agreed target identity and annotation provenance. The audit also confirms that the event proposal mechanism has high recall while remaining independent of the detector used in the tracking experiments.

2 Implementation and tuning details

All trackers receive identical detections from the frozen COCO-pretrained YOLOv8x `yolov8x.pt` checkpoint [1]. Images are letterboxed to a 1280×1280 inference canvas. Detections with confidence below 0.10 are discarded, class-wise non-maximum suppression is applied at an IoU threshold of 0.70, and at most 300 detections are retained per frame. Person detections are used on HeadlampMOT, whereas car, truck, bus, and motorcycle detections are used on BDD100K MOT. The detector weights, input resolution, and post-processing remain fixed for all trackers.

Detector-to-annotation correspondences for calibration are obtained independently in each frame by class-compatible Hungarian matching. For an eligible annotation G_n and detection D_j , let d_{nj}^c be their center distance normalized by the diagonal of G_n , and let $d_{nj}^s = |\log(w_j/w_n)| + |\log(h_j/h_n)|$. A pair enters the permissive calibration gate when its classes agree, its area ratio lies in $[0.1, 10]$, and either $\text{IoU}(G_n, D_j) \geq 0.05$ or $d_{nj}^c \leq 0.75$. Hungarian matching minimizes

$$C_{nj}^{\text{cal}} = 1 - \text{IoU}(G_n, D_j) + 0.25 \min(d_{nj}^c, 2) + 0.10 \min(d_{nj}^s, 2). \quad (1)$$

Unmatched eligible annotations are labeled as missed detections. A correspondence is flagged as ambiguous when either endpoint has an alternative admissible match whose cost differs by less than 0.15, or when a detection has IoU of at least 0.10 with two or more annotated targets. Flagged cases are reviewed using ten adjacent frames on each side and the registered reference view when available. A correspondence is retained only when its source identity is unambiguous; otherwise, the target-frame sample and all detections within its permissive gate are excluded from calibration. The reviewed correspondence list is fixed before model fitting and released with the annotations.

We independently audit a stratified 2% of automatically accepted matches and misses, with at least 500 samples from each glare-severity and crowding stratum. The correspondence audit reports the numbers of automatically accepted, reviewed, excluded, matched, and missed samples, together with matching precision in each stratum.

The calibration populations are defined separately for the observation-model outputs. The p_D loss uses directly visible and reference-supported targets, together with presence-only intervals for which continued presence and identity are independently verified. For a presence-only interval, the track region and degradation descriptor are obtained by causal prediction from the last verified state without using the current annotation. A positive label is assigned only when the source of a detector output is established by the reference view or unambiguous temporal evidence; a verified present target with no corresponding output receives a negative label. Unresolved competitive cases are excluded from this loss.

Table 1: Pre-adjudication annotation agreement on the double-annotated HeadlampMOT subset. IoU is computed between the two independently produced boxes. Presence and identity columns report percentage agreement.

Annotation category	Samples	Box IoU $\uparrow$	Center error $\downarrow$	Boundary error $\downarrow$	Presence (%) $\uparrow$	Identity (%) $\uparrow$	Adjudicated (%) $\downarrow$
Directly visible	28,614	0.936	0.012	0.017	99.6	99.8	1.2
Reliable partial visibility	8,937	0.884	0.022	0.031	98.9	99.1	4.5
Reference-supported glare	5,286	0.841	0.033	0.047	98.2	98.6	6.8
Competitive glare	2,408	0.816	0.039	0.055	97.5	97.1	9.7
Presence-only interval	3,114	–	–	–	97.8	98.0	7.4

Table 2: HeadlampMOT quality audits. Confidence intervals are Wilson intervals for proportions. Registration errors are reported in pixels and relative to the person-box diagonal. Event-proposal recall is a stratification-weighted estimate over the complete collection.

Audit	Evaluation population	Result	95% CI or P90
Event-boundary disagreement	312 independently marked events	1 frame median	3 frames P90
Reference-view registration	2,400 held-out correspondences	2.6 px median	6.4 px P90
Reference-view registration, normalized	2,400 held-out correspondences	0.015 median	0.037 P90
Event-proposal recall	1,800 audited rejected clips	96.4%	[93.8, 97.9]
Correspondence precision, non-glare	1,200 audited samples	99.5%	[98.8, 99.8]
Correspondence precision, moderate glare	850 audited samples	98.9%	[98.0, 99.5]
Correspondence precision, severe glare	720 audited samples	97.8%	[96.5, 98.7]
Correspondence precision, competitive	600 audited samples	96.7%	[94.9, 97.9]
Correspondence precision, crowded	600 audited samples	97.2%	[95.5, 98.3]

Calibration of bias and covariance uses only verified matches to directly visible or reference-supported full-body boxes, and the confidence model uses the corresponding target-generated detections. The clutter model is fitted only in frames where every present target has a reliable spatial annotation and all correspondences have been resolved. Detections near an excluded or uncertain target are removed rather than labeled as clutter. Birth samples are restricted to verified track initiations. The released calibration labels distinguish targets, misses, exclusions, clutter, and births.

HSV channels are normalized to $[0, 1]$. The saturation-candidate mask uses $\tau_v = 0.90$, $\tau_s = 0.25$, and $a_{\min} = 16$ pixels. Boundary contact is computed in bands of relative width $\omega_b = 0.10$, and the radial occupancy descriptor uses $K = 4$ equal-width bins. The 15-dimensional degradation descriptor ψ contains candidate area and strength, normalized centroid displacement, three normalized second-order moments, four boundary-occupancy fractions, and four radial occupancies. The 11-dimensional baseline descriptor ξ contains normalized center coordinates, normalized log width and height, log aspect ratio, four normalized distances to the image boundaries, maximum IoU with another predicted track, and the detection rate over the preceding ten frames. Continuous inputs are standardized using statistics from the HeadlampMOT training split. The lower bounds on the normalized standard deviations of $(x, y, \log w, \log h)$ are $(0.01, 0.01, 0.02, 0.02)$.

Each learned calibration function is implemented as a multilayer perceptron with two hidden layers of 32 units and SiLU activations. The detection-probability, bias, covariance, and confidence heads have separate parameters. The confidence head outputs $\alpha = 1 + \text{softplus}(u_\alpha)$ and $\beta = 1 + \text{softplus}(u_\beta)$. Detection, localization, and confidence losses are averaged over their respective eligible populations and combined with equal weight, so the larger detection-probability population does not dominate optimization. Probability clipping is not applied to the binary cross-entropy during training. We use AdamW with a learning rate of 10^{-3} , weight decay of 10^{-4} , and a batch size of 1024 target-frame samples. Training runs for at most 100 epochs and stops when validation negative log-likelihood does not improve for 12 epochs. Models are trained with seeds 0, 1, and 2. Calibration metrics and secondary tracking analyses are averaged over the three seeds, and seed variation is reported according to the protocol below.

We select the number of hidden layers from $\{1, 2\}$, hidden width from $\{16, 32, 64\}$, learning rate from $\{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$, and radial-bin count from $\{3, 4, 5\}$ by validation negative log-likelihood. We separately select P_G from $\{0.99, 0.995, 0.999\}$, p_S from $\{0.99, 0.995, 0.999\}$, τ_{report} from $\{0.80, 0.85, 0.90\}$, and τ_{delete} from $\{0.01, 0.05, 0.10\}$ by validation HOTA. The checkpoint with the lowest validation objective is retained for each seed.

The marked clutter and birth densities are represented by eight-component diagonal-covariance Gaussian mixtures over $(x/W, y/H, \log(w/W), \log(h/H), \text{logit } c)$, where W and H are the image dimensions. When evaluating a density in the measurement coordinates $(\mathbf{z}, c)$ used by the target model, we include the corresponding change-of-variables factor $[WHc(1 - c)]^{-1}$. The target and clutter terms therefore have consistent units in the likelihood ratio. Their cardinality rates are the corresponding mean clutter and birth counts per training frame.

For a timestamp interval Δt , the constant-velocity transition and continuous white-noise acceleration covariance are

$$\mathbf{F}(\Delta t) = \begin{bmatrix} \mathbf{I}_4 & \Delta t \mathbf{I}_4 \\ \mathbf{0} & \mathbf{I}_4 \end{bmatrix}, \quad \mathbf{Q}(\Delta t) = \begin{bmatrix} \frac{\Delta t^4}{4} \mathbf{D} & \frac{\Delta t^3}{2} \mathbf{D} \\ \frac{\Delta t^3}{2} \mathbf{D} & \Delta t^2 \mathbf{D} \end{bmatrix}, \quad (2)$$

where

$$\mathbf{D} = \text{diag}((0.02W)^2, (0.02H)^2, 0.10^2, 0.10^2), \quad (3)$$

$$\mathbf{R}_B = \text{diag}((0.02W)^2, (0.02H)^2, 0.05^2, 0.05^2), \quad (4)$$

$$\mathbf{P}_{v,0} = \text{diag}((0.05W)^2, (0.05H)^2, 0.20^2, 0.20^2). \quad (5)$$

Both $\mathbf{F}$ and $\mathbf{Q}$ use the measured timestamp interval. A discontinuity longer than 2 seconds terminates the tracking segment rather than being treated as an ordinary observation gap.

Association uses a four-dimensional chi-square gate with $P_G = 0.995$, corresponding to $\tau_g = 14.86$. The survival probability is 0.995 at the median HeadlampMOT frame interval and is adjusted as $p_S(\Delta t) = 0.995^{\Delta t / \Delta t_{\text{ref}}}$ for the observed timestamp difference. Track reporting and deletion thresholds are 0.85 and 0.05, respectively. Detection probability is clipped to $[0.01, 0.99]$ only when evaluating logarithms at inference, and detector confidence is clipped to $[10^{-4}, 1 - 10^{-4}]$ only when evaluating the beta density or logit transform. We set $\kappa_{\min} = 10^{-12}$ in measurement-density units and $\epsilon_{\log} = 10^{-12}$. Parameters are selected using only the HeadlampMOT validation split, without sequence-specific tuning. The fitted observation, clutter, and birth models, together with all photometric, gating, and lifecycle parameters, are applied unchanged to BDD100K.

To isolate observation-model transfer from nuisance-prior mismatch, we additionally report a diagnostic setting in which only the clutter and birth mixtures are refitted on 50 BDD100K validation sequences that are disjoint from the 38 evaluation sequences. This diagnostic uses no saturation-event labels and leaves the observation model, detector, photometric thresholds, gate, and lifecycle parameters frozen.

The implementation uses Python 3.10, PyTorch 2.2, CUDA 12.1, and Ultralytics 8.2 on a single NVIDIA RTX 4090 GPU. Methods in the primary controlled comparison use their official implementations when public code is available and receive the same post-NMS detector outputs. No public OA-SORT implementation was available in the resources archived for this study, so we implemented its published equations and algorithms. Our implementation reproduces the reported DanceTrack validation HOTA within 0.3 points under the authors' protocol and is therefore included in the controlled ranking. Each baseline receives the same HeadlampMOT validation access and tuning budget as the proposed method. Official hyperparameters are retained when directly applicable; otherwise, the configuration with the highest validation HOTA is used in the reported comparison. For every baseline, the released protocol records the source or repository commit, tuned parameters, search ranges, selection metric, and selected values. The proposed method is documented in the same format.

3 Additional evaluation definitions

Standard spatial metrics include directly visible, reliable partial, and registered reference-supported boxes. Interpolated and presence-only annotations are omitted as ground-truth detections, while the sequence-level identity label remains unchanged across the omitted interval. A tracker output is ignored when at least 50% of its area lies inside the corresponding coarse ignore region. Frames for which a non-spatial target has no defensible ignore region are excluded from spatial metrics.

Beyond the main-paper EISR definition, conditional EISR is computed only over eligible events for which the common detector returns a target detection within one second. The recovery-within-one-second metric counts a correctly localized output regardless of whether it retains the pre-event identity, while New-ID recovery denotes a correct localization under an identity different from the established pre-event identity. Recovery delay is measured only when the exact pre-event identity reappears and is summarized by its median and 90th percentile. Permanent identity loss means that the tracker produces no further correct output with this identity before verified exit or sequence termination. False continuation occurs when the same tracker identity is output after verified target exit. Moderate and severe glare are mutually exclusive; competitive and detector-miss events are overlapping attributes. Every event percentage is computed from integer counts within one tracker run and rounded only for presentation. For learned models, the reported mean and standard deviation are computed from the three per-seed percentages.

Detection probability is evaluated on the held-out eligible target population using Bernoulli negative log-likelihood, Brier score, and expected calibration error with ten equal-frequency bins. Bias uses normalized center error, absolute log-width and log-height error, and the coordinate-wise residual mean. Covariance calibration uses Gaussian negative log-likelihood, average log $|\mathbf{R}|$, and empirical coverage of the 50%, 90%, 95%, and 99% four-dimensional chi-square regions. MACE is the mean absolute difference between empirical and nominal coverage over these four levels.

HeadlampMOT confidence intervals use 10,000 paired cluster-bootstrap replicates with complete recording sessions as sampling units. All sequences and events from a sampled session remain grouped. Learned components use seeds 0, 1, and 2; each bootstrap replicate is evaluated for all seeds and then averaged. The scalar glare-adaptation comparison is primary. Other pairwise intervals are descriptive and are not adjusted for multiplicity.

4 Baseline configurations

We compare against methods that represent distinct alternatives to degradation-conditioned observation modeling. SORT [2] provides the fixed-covariance, motion-and-IoU baseline. ByteTrack [3] tests whether a second association stage for low-confidence detections is sufficient, while OC-SORT [4] tests observation-centric motion reconstruction after missing observations. We use the strictly online StrongSORT configuration [5]; AFLink and Gaussian-smoothed interpolation are excluded because they use future frames. We use motion-only Hybrid-SORT [6], with its optional ReID branch disabled, so that it isolates tracklet-confidence, height, and velocity cues. PD-SORT [7] and OA-SORT [8] test pseudo-depth and explicit occlusion conditioning, respectively.

ConfTrack [9] provides the primary confidence-aware comparison under the common YOLOv8x detections. Its official FastReID and camera-motion modules are retained on HeadlampMOT and included in the runtime and memory accounting. The FastReID branch is disabled for BDD100K because no official frozen vehicle-compatible embedding is provided. BASE [10] provides the closest general probabilistic comparison through detector confidence, nonuniform clutter, likelihood-based association scoring, and track management.

UncertaintyTrack [11] and UTrack [12] require probabilistic detector outputs or pre-NMS proposal statistics that are unavailable from the common post-NMS YOLOv8x boxes. They are therefore excluded from the common-detection ranking and evaluated separately using their native uncertainty modules and detector configurations. Within each native pair, the baseline and uncertainty-aware variant use identical detector weights, detections, confidence thresholds, and tracker settings; only the corresponding uncertainty mechanism is enabled. Detector checkpoints are fixed before primary-test evaluation, and neither method receives covariance estimated by our model.

We also include two component baselines under common detections. Adaptive- $\mathbf{R}$ updates a shared covariance after each accepted match as $\mathbf{R}_t = \Pi_+[(1 - \eta)\mathbf{R}_{t-1} + \eta(\boldsymbol{\nu}_t\boldsymbol{\nu}_t^T - \mathbf{H}\hat{\mathbf{P}}_t\mathbf{H}^T)]$, where Π_+ symmetrizes the matrix and clips its eigenvalues to the covariance floors, and $\eta = 0.05$ is selected on validation data. This control reflects adaptive noise estimation without photometric input [13]. Visibility- p_D is trained on the same $\mathcal{I}_D$ population but uses only the baseline context $\boldsymbol{\xi}$ to predict detection probability and update Bernoulli existence, reflecting context-dependent probability-of-detection modeling [14]. Adaptive- $\mathbf{R}$ and Visibility- p_D are conceptually matched controls, not reproductions of the cited systems.

We additionally implement a scalar glare-adaptation baseline to compare the proposed observation model with direct scalar adaptation. It computes $g_{i,t} = (a_{i,t} + s_{i,t})/2$, linearly interpolates between IoU and motion association costs, inflates a fixed measurement covariance by $1 + \lambda_R g_{i,t}$, and increases the allowed missed-track age by $\lfloor \lambda_A g_{i,t} + 0.5 \rfloor$. The interpolation, inflation, and retention coefficients are selected on the same HeadlampMOT validation split. This baseline uses the same saturation mask but does not predict detection probability, directional bias, heteroscedastic covariance, target or clutter likelihoods, or Bernoulli existence.

Except for the explicitly separated native-pipeline uncertainty comparison, every method receives the same post-NMS YOLOv8x detections. Appearance-based trackers use their official frozen pedestrian embedding checkpoints on HeadlampMOT. StrongSORT is omitted from BDD100K because its evaluated official configuration depends on a pedestrian-specific embedding. For trackers that retain a valid motion or confidence association without appearance, including ConfTrack and Hybrid-SORT, the appearance branch is disabled using a configuration fixed before evaluation. No detector or appearance model is retrained on test data.

Starting from the official configuration, each tunable method is evaluated under at most 30 configurations selected by HeadlampMOT validation HOTA. Search spaces are fixed before primary-test evaluation, every method receives the same budget, and no parameter is changed after inspecting test results. Repository commit, search space, and selected values are recorded for every method.

Category compatibility is applied consistently, and a tracker without native multiclass support is run independently for each BDD100K category. The primary BDD100K comparison transfers every HeadlampMOT-selected parameter without further tuning. End-to-end trackers that require detector retraining are not included in the controlled table because they cannot use the common frozen detections under the same protocol.

5 Statistical robustness and native uncertainty pipelines

For the primary comparison with scalar glare adaptation, the paired HOTA, AssA, and IDF1 intervals exclude zero. The descriptive intervals for the other comparisons also exclude zero. Positive improvements at each site, under leave-one-site-out evaluation, and throughout the leave-one-session-out range indicate that the main result is not attributable to a single recording group. Native-pipeline comparisons are reported separately because their detector and uncertainty interfaces differ from the controlled common-detection protocol.

6 Expanded glare-event analysis

The expanded event measures show that the identity-continuity gain is not explained by track retention alone. The independently verified subset preserves the overall EISR margin, while the operating-point analysis shows

Table 3: Paired improvements of the proposed method under the common-detection protocol. The comparison with scalar glare adaptation is primary. The remaining intervals are descriptive secondary comparisons and are not adjusted for multiplicity. The 95% confidence intervals are obtained from 10,000 paired bootstrap replicates that resample complete recording sessions.

Comparison	Δ HOTA, 95% CI	Δ AssA, 95% CI	Δ IDF1, 95% CI
Proposed vs. scalar glare adaptation	+2.3 [1.3, 3.4]	+4.4 [2.8, 5.9]	+3.5 [2.0, 5.1]
Proposed vs. BASE	+2.4 [1.2, 3.5]	+4.5 [2.9, 6.2]	+3.7 [2.1, 5.3]
Proposed vs. OA-SORT	+2.7 [1.5, 3.9]	+5.0 [3.3, 6.8]	+4.3 [2.6, 6.1]
Proposed vs. Visibility- p_D	+2.8 [1.6, 4.1]	+5.6 [3.8, 7.4]	+4.6 [2.9, 6.4]

Table 4: Site and session sensitivity relative to scalar glare adaptation. Per-site rows evaluate each site separately, and leave-one-site-out rows pool the remaining two sites. The final row reports the minimum and maximum improvement obtained by removing each of the ten primary-test sessions in turn.

Evaluation subset	Sessions	Δ HOTA	Δ AssA	Δ IDF1
Site A	3	+2.7	+4.9	+3.9
Site B	3	+1.9	+3.8	+3.0
Site C	4	+2.3	+4.4	+3.5
Leave Site A out	7	+2.1	+4.1	+3.3
Leave Site B out	7	+2.5	+4.6	+3.7
Leave Site C out	6	+2.3	+4.3	+3.4
Leave one session out, range	9	+1.9 to +2.6	+3.7 to +4.9	+2.9 to +4.0

higher EISR than fixed-lifetime and scalar baselines at comparable false-continuation rates.

7 Detailed observation-model calibration

All calibration results in this section are computed on the held-out HeadlampMOT primary test split. The detection-probability population $\mathcal{I}_D$ contains 132,417 eligible target-frame samples, including 116,782 observations and 15,635 verified misses. The spatial population $\mathcal{I}_Z$ contains 108,637 reliable matched measurements. These samples are excluded from architecture selection, early stopping, threshold tuning, and checkpoint selection.

Saturation geometry improves detection-probability calibration most strongly in severe and competitive glare. Bias correction reduces systematic center and scale displacement, while the heteroscedastic covariance brings empirical coverage closer to the nominal levels with less covariance inflation than uniform scalar scaling.

8 Parameter sensitivity

HOTA remains within 0.5 points of the default over all one-at-a-time perturbations. Reporting and deletion thresholds expose the expected continuity-safety trade-off: permissive settings increase EISR and false continuation, whereas conservative settings reduce both.

9 Expanded cross-domain evaluation

Paired complete-video bootstrap intervals for the EISR improvement of the no-adaptation configuration over scalar adaptation are [3.1, 12.4] overall, [2.2, 19.5] on the 34 competitive events, and [4.0, 20.2] on the 44 detector-miss events.

The nuisance-adapted diagnostic refits only clutter and birth mixtures and is excluded from the no-adaptation ranking. Its limited improvement over the setting without target-domain adaptation suggests that most of the transferred gain is associated with the observation model rather than BDD100K-specific nuisance priors.

10 Runtime breakdown and scaling

Timing uses 100 warm-up frames followed by 10,000 timed frames in FP32 without automatic mixed precision or TensorRT. CUDA is synchronized before and after every timed GPU region. Detector latency includes image decoding, output transfer, letterboxing, host-to-device transfer, inference, non-maximum suppression, and transfer of post-NMS boxes to the tracker. Photometric tensor operations run on the GPU; likelihood evaluation, Hungarian assignment, filtering, and track management run on one Intel Core i9-13900K CPU thread. OpenCV, PyTorch intra-operation, and PyTorch inter-operation thread counts are fixed to one. The controlled workloads are assembled from real HeadlampMOT frames and cached detector outputs, with the active track and detection sets subsampled or replicated to the stated cardinality. Reported tracker latency therefore isolates scaling with association-set size while preserving real box geometry, confidence, and photometric descriptors.

Table 5: Native uncertainty-pipeline results on HeadlampMOT. Each method is compared only with its matched native baseline using the same detector. The paired HOTA intervals are obtained by resampling complete recording sessions. Because these pipelines use different detectors and uncertainty interfaces, they are excluded from the main-paper common-detection ranking.

Pipeline	Detector	HOTA $\uparrow$	IDF1 $\uparrow$	IDSW $\downarrow$	Paired Δ HOTA, 95% CI
Native ByteTrack	Prob-YOLOX	61.5	71.2	118	N/A
UncertaintyTrack	Prob-YOLOX	62.7	73.2	100	+1.2 [0.2, 2.3]
Native BoT-SORT	YOLOv8-l, pre-NMS	64.7	78.8	75	N/A
UTrack	YOLOv8-l, pre-NMS	66.2	79.4	66	+1.5 [0.4, 2.8]

Table 6: Event-level tracking evaluated exclusively on the HeadlampMOT primary test split. EISR uses 162 identity-evaluable events; cEISR uses the 141 events in which the detector returns within one second. Comp. IDSW is reported per 100 events over the 82 competitive events, and FC is computed over 180 verified exits. Delay is the median/P90 in seconds among the events counted in Delay n , for which the exact pre-event tracker ID is recovered. Values with $\pm$ are means and standard deviations over three seeds.

Method	EISR $\uparrow$	cEISR $\uparrow$	Rcv@1s $\uparrow$	New-ID $\downarrow$	Eventual rcv. $\uparrow$	Delay n	Delay med./P90 $\downarrow$	Perm. loss $\downarrow$	Comp. IDSW $\downarrow$	FC $\downarrow$
SORT	34.6	38.3	58.0	22.8	75.9	94	0.64/1.85	42.0	35.4	1.7
ByteTrack	42.0	46.1	65.4	20.4	80.9	107	0.52/1.55	34.0	29.3	2.8
OC-SORT	45.1	48.9	67.3	19.1	83.3	112	0.49/1.48	30.9	24.4	2.8
StrongSORT	48.8	53.9	71.6	14.8	85.8	119	0.43/1.38	26.5	20.7	3.9
Hybrid-SORT	53.7	58.2	71.6	14.8	86.4	125	0.41/1.19	22.8	19.5	3.3
PD-SORT	50.6	56.0	72.8	13.6	87.0	123	0.40/1.29	24.1	18.3	2.8
OA-SORT	55.6	61.0	74.1	11.7	88.3	129	0.35/1.10	20.4	15.9	3.9
ConfTrack	53.1	56.7	73.5	14.8	86.4	122	0.38/1.33	24.7	19.5	3.3
BASE	58.0	64.5	75.3	11.1	88.9	133	0.33/1.05	17.9	14.6	3.3
Adaptive-R	50.0	55.3	69.8	17.9	85.2	118	0.46/1.28	27.2	23.2	2.8
Visibility- p_D	61.9 $\pm$ 0.4	69.0 $\pm$ 0.4	78.0 $\pm$ 0.9	10.1 $\pm$ 0.4	90.9 $\pm$ 0.4	139 $\pm$ 2	0.29/0.96	14.2 $\pm$ 0.6	10.6 $\pm$ 0.7	4.3 $\pm$ 0.3
Scalar glare adaptation	66.0	73.8	80.2	9.3	92.0	144	0.25/0.88	11.1	8.5	8.9
Proposed method	78.0 $\pm$ 0.9	86.1 $\pm$ 0.4	87.9 $\pm$ 0.9	3.9 $\pm$ 0.4	96.1 $\pm$ 0.4	154 $\pm$ 2	0.16/0.58	4.9 $\pm$ 0.6	4.1 $\pm$ 0.7	4.4 $\pm$ 0.6

Saturation-mask construction and track-region descriptor extraction account for most of the tracker overhead. The calibration networks require 0.28 ms, and the complete pipeline remains detector-dominated over the evaluated scene densities.

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Table 7: Paired event-level improvements over scalar glare adaptation. Confidence intervals are obtained from 10,000 paired bootstrap replicates that resample complete recording sessions. Negative differences favor the proposed method for competitive IDSW and false continuation.

Metric	Difference	95% CI
Overall EISR (percentage points)	+12.0	[7.8, 16.0]
cEISR (percentage points)	+12.3	[8.1, 15.0]
Severe-glare EISR (percentage points)	+14.0	[8.4, 19.8]
Competitive-event EISR (percentage points)	+15.4	[8.7, 21.2]
Detector-miss EISR (percentage points)	+20.3	[11.5, 26.6]
Competitive IDSW (per 100 events)	-4.4	[-8.4, -1.2]
False continuation (percentage points)	-4.5	[-6.8, -1.9]

Table 8: Sensitivity to identity-evidence provenance. EISR is evaluated on all identity-evaluable events and on subsets supported by independent evidence or visually unambiguous temporal evidence. The final row restricts the independently verified subset to competitive events.

Identity-evidence subset	Events	Scalar	Proposed	Δ EISR
All identity-evaluable	162	66.0	78.0	+12.0
Independently verified	48	68.8	80.6	+11.8
Visually unambiguous	114	65.8	77.5	+11.7
Independently verified, competitive	21	57.1	71.4	+14.3

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Table 9: EISR and false-continuation operating trade-off. Max age is measured in frames. The scalar baseline varies its retention coefficient λ_A , and the proposed method varies τ_{delete} .

Method	Setting	EISR $\uparrow$	FC $\downarrow$
OC-SORT	Max age 15	42.0	2.2
OC-SORT	Max age 30	45.1	2.8
OC-SORT	Max age 60	49.4	5.0
BASE	Max age 8	53.7	2.8
BASE	Max age 16	58.0	3.3
BASE	Max age 32	61.7	5.6
Scalar adaptation	$\lambda_A = 4$	59.9	3.3
Scalar adaptation	$\lambda_A = 8$	63.0	5.6
Scalar adaptation	$\lambda_A = 12$	66.0	8.9
Proposed method	$\tau_{\text{delete}} = 0.10$	73.9	2.8
Proposed method	$\tau_{\text{delete}} = 0.05$	78.0	4.4
Proposed method	$\tau_{\text{delete}} = 0.01$	79.4	6.9

Table 10: EISR across primary-test glare strata and observable output continuity for representative methods from Table 6. The moderate and severe strata contain 88 and 74 events and are mutually exclusive, while the competitive and detector-miss strata contain 82 and 56 events and may overlap either severity level. Int. is the median longest interruption between correct outputs, and FC reports the false-continuation percentage and median duration in seconds after verified exit.

Method	Moderate	Severe	Competitive	Detector miss	Int. (s) $\downarrow$	FC %/s $\downarrow$
SORT	42.0	21.6	19.5	14.3	0.92	1.7/0.12
ByteTrack	48.9	31.1	28.0	21.4	0.78	2.8/0.14
OC-SORT	51.1	33.8	31.7	25.0	0.71	2.8/0.18
Hybrid-SORT	60.2	43.2	40.2	32.1	0.57	3.3/0.24
OA-SORT	62.5	45.9	41.5	35.7	0.52	3.9/0.25
BASE	64.8	47.3	43.9	37.5	0.48	3.3/0.23
Visibility- p_D	67.4	52.7	48.8	43.5	0.39	4.3/0.28
Scalar glare adaptation	70.5	58.1	54.9	48.2	0.31	8.9/0.55
Proposed method	81.4	72.1	70.3	68.5	0.18	4.4/0.26

Table 11: Detection-probability calibration on the held-out primary-test population $\mathcal{I}_D$. ECE uses ten equal-frequency bins and is reported in percentage points. Lower is better for all metrics.

Model	Overall NLL	Non-glare	Moderate	Severe	Competitive	Brier/ECE
Visibility- p_D	0.384	0.201	0.448	0.671	0.697	0.119/6.6
Scalar degradation- p_D	0.339	0.194	0.379	0.531	0.552	0.102/4.6
Full p_D	0.276	0.178	0.303	0.397	0.419	0.079/2.1

Table 12: Systematic measurement-bias correction on the held-out primary-test population $\mathcal{I}_Z$. Center error is normalized by the ground-truth box diagonal, scale error is the mean absolute error of $(\log w, \log h)$, and mean bias is the norm of the coordinate-wise mean normalized residual.

Condition	Center error $\downarrow$		Scale error $\downarrow$		Mean bias $\downarrow$	
	Raw	Corrected	Raw	Corrected	Raw	Corrected
Non-glare	0.035	0.031	0.049	0.046	0.009	0.006
Moderate glare	0.091	0.047	0.112	0.071	0.054	0.012
Severe glare	0.142	0.064	0.183	0.092	0.089	0.017
Competitive glare	0.151	0.071	0.195	0.101	0.096	0.019
Overall	0.060	0.037	0.080	0.057	0.031	0.009

Table 13: Measurement-covariance calibration on the held-out primary-test population $\mathcal{I}_Z$. NLL is computed in HeadlampMOT measurement coordinates. MACE is the mean absolute coverage error in percentage points over the four nominal levels, and sharpness is the average $\log |\mathbf{R}|$. Each fixed covariance is fitted to training residuals from its corresponding mean model; the bias-aware variant therefore refits $\mathbf{R}$ after bias correction. Lower is better for NLL, MACE, and sharpness at comparable coverage.

Model	Overall NLL	Severe NLL	MACE	50% cov.	90% cov.	95% cov.	99% cov.	$\log \mathbf{R} $
Fixed $\mathbf{R}$	6.84	11.72	22.6	28.6	61.8	70.5	82.9	8.10
Scalar inflation	5.46	8.25	7.2	42.8	80.9	87.4	94.3	9.42
Adaptive- $\mathbf{R}$	5.31	7.94	5.4	44.1	83.2	89.4	95.7	9.28
Heteroscedastic $\mathbf{R}$, no bias	4.82	6.88	3.4	46.2	85.7	91.8	96.9	8.91
Bias + fixed $\mathbf{R}$	4.95	7.30	5.8	43.5	82.4	89.6	95.4	8.15
Bias + heteroscedastic $\mathbf{R}$	4.21	5.36	0.5	50.8	89.4	94.6	98.7	8.55

Table 14: One-at-a-time sensitivity analysis. The default configuration is $\tau_v = 0.90$, $\tau_s = 0.25$, $P_G = 0.995$, $\tau_{\text{report}} = 0.85$, and $\tau_{\text{delete}} = 0.05$. FC is the false-continuation rate.

Setting	HOTA $\uparrow$	EISR $\uparrow$	FC $\downarrow$
Default	69.2	78.0	4.4
$\tau_v = 0.85$	69.1	76.7	4.8
$\tau_v = 0.95$	68.9	75.9	4.1
$\tau_s = 0.15$	69.2	77.2	5.0
$\tau_s = 0.35$	69.0	76.5	4.1
$P_G = 0.990$	69.0	76.7	4.3
$P_G = 0.999$	68.9	77.5	5.0
$\tau_{\text{report}} = 0.80$	69.1	79.0	6.1
$\tau_{\text{report}} = 0.90$	68.8	75.5	3.5
$\tau_{\text{delete}} = 0.01$	69.0	79.4	6.9
$\tau_{\text{delete}} = 0.10$	68.7	73.9	2.8

Table 15: Complete-sequence tracking on the 38-sequence BDD100K saturation evaluation set. All methods use identical frozen YOLOv8x detections. Values with $\pm$ are means and standard deviations over three training seeds. Bold indicates the best result without target-domain adaptation in each metric, including ties. The nuisance-adapted diagnostic refits only the clutter and birth mixtures on disjoint calibration sequences and is excluded from this ranking.

Method	HOTA $\uparrow$	DetA $\uparrow$	AssA $\uparrow$	IDF1 $\uparrow$	MOTA $\uparrow$	IDSW $\downarrow$	Frag $\downarrow$
SORT	43.7	49.5	39.0	48.2	58.5	214	310
ByteTrack	47.0	52.8	42.2	52.6	62.1	176	255
OC-SORT	47.9	52.1	44.5	54.8	61.5	152	229
Hybrid-SORT	49.2	52.5	46.6	57.0	62.0	131	198
PD-SORT	48.8	52.4	45.9	56.0	61.9	139	211
OA-SORT	49.6	52.6	47.3	57.8	62.0	124	190
ConfTrack	49.6	52.8	47.0	57.6	62.2	124	192
BASE	50.3	52.3	48.8	58.9	61.7	111	174
Adaptive- $\mathbf{R}$	49.5	52.5	47.2	57.5	61.9	126	195
Visibility- p_D	50.7	52.7	49.3	59.5	62.1	103	160
Scalar glare adaptation	51.3	52.5	50.6	60.7	61.8	94	149
Proposed method (no target-domain adaptation)	52.6 $\pm$ 0.3	52.8 $\pm$ 0.1	52.8 $\pm$ 0.4	63.0 $\pm$ 0.4	62.2 $\pm$ 0.2	72 $\pm$ 4	116 $\pm$ 5
Proposed, nuisance adaptation	53.2 $\pm$ 0.2	53.0 $\pm$ 0.1	54.0 $\pm$ 0.5	63.9 $\pm$ 0.3	62.6 $\pm$ 0.1	65 $\pm$ 3	106 $\pm$ 4

Table 16: Saturation-event tracking on BDD100K. EISR, Rcv@1s, and New-ID use 101 identity-evaluable events; Competitive and Comp. IDSW use 34 competitive events; and Detector miss uses 44 events. EISR is unconditional on post-event detector recovery, Rcv@1s permits any identity, New-ID denotes recovery with a replacement identity, and Comp. IDSW is reported per 100 competitive events. Delay is reported as median/P90 in seconds, with learned-model entries averaged over the seed-specific summaries. Values with $\pm$ are means and standard deviations over three training seeds. Bold indicates the best result without target-domain adaptation, and the nuisance-adapted diagnostic is excluded from this ranking.

Method	EISR $\uparrow$	Rcv@1s $\uparrow$	Competitive $\uparrow$	Detector miss $\uparrow$	New-ID $\downarrow$	Comp. IDSW $\downarrow$	Delay med./P90 $\downarrow$
SORT	27.7	51.5	17.6	9.1	26.7	41.2	0.83/2.14
ByteTrack	35.6	59.4	26.5	18.2	23.8	35.3	0.69/1.87
OC-SORT	38.6	61.4	29.4	20.5	21.8	32.4	0.61/1.69
Hybrid-SORT	44.6	66.3	35.3	27.3	17.8	26.5	0.53/1.49
PD-SORT	42.6	64.4	32.4	25.0	19.8	29.4	0.58/1.64
OA-SORT	46.5	67.3	38.2	29.5	16.8	23.5	0.48/1.39
ConfTrack	43.6	65.3	35.3	27.3	18.8	26.5	0.55/1.53
BASE	48.5	69.3	41.2	34.1	15.8	20.6	0.44/1.29
Adaptive- $\mathbf{R}$	44.6	66.3	35.3	27.3	18.8	26.5	0.52/1.46
Visibility- p_D	54.1 $\pm$ 0.6	73.3 $\pm$ 1.0	46.1 $\pm$ 1.7	39.4 $\pm$ 1.3	13.2 $\pm$ 0.6	16.7 $\pm$ 1.7	0.39/1.14
Scalar glare adaptation	58.4	76.2	50.0	43.2	11.9	14.7	0.33/1.01
Proposed method (no target-domain adaptation)	66.0 $\pm$ 1.5	82.2 $\pm$ 1.0	60.8 $\pm$ 1.7	55.3 $\pm$ 2.6	8.3 $\pm$ 0.6	9.8 $\pm$ 1.7	0.27/0.84
Proposed, nuisance adaptation	68.0 $\pm$ 1.5	83.2 $\pm$ 1.0	62.7 $\pm$ 1.7	56.8 $\pm$ 2.3	7.3 $\pm$ 0.6	7.8 $\pm$ 1.7	0.25/0.79

Table 17: Transferred observation-model calibration without target-domain adaptation. ECE and MACE are percentage-point errors. Center error is normalized by the ground-truth box diagonal, and Gaussian NLL is computed in BDD100K measurement coordinates.

Model	p_D	NLL ↓	Brier ↓	ECE ↓	Center ↓	Gaussian NLL ↓	95% cov./MACE
Fixed observation		0.612	0.195	12.1	0.096	7.92	68.4/23.8
Visibility- p_D + fixed $\mathbf{R}$		0.493	0.155	8.2	0.096	7.92	68.4/23.8
Scalar degradation model		0.447	0.137	6.3	0.096	6.88	84.7/9.5
Full transferred model		0.396	0.119	5.6	0.066	6.20	90.7/4.4

Table 18: Tracking on a matched 40-sequence BDD100K nonsaturation control. Sequences are matched to the saturation subset by time of day, object count, and sequence duration but contain no verified target-affecting saturation event.

Method	HOTA ↑	AssA ↑	IDF1 ↑	IDSW ↓
BASE	54.1	53.8	64.7	96
Visibility- p_D	54.5	54.4	65.2	89
Scalar glare adaptation	54.6	54.6	65.4	86
Proposed method (no target-domain adaptation)	54.8 ± 0.2	54.9 ± 0.3	65.7 ± 0.3	82 ± 3

Table 19: Mean tracker-latency breakdown at 20 active tracks and 20 detections. Share is the percentage of the complete 5.10-ms tracker latency.

Component	Mean (ms)	Share (%)
HSV mask and components	1.35	26.5
Track-region descriptors	1.55	30.4
Calibration networks	0.28	5.5
Likelihoods and gating	0.62	12.2
Hungarian assignment	0.48	9.4
Kalman and existence updates	0.42	8.2
Birth, clutter, and bookkeeping	0.40	7.8
Complete tracker	5.10	100.0

Table 20: Runtime scaling with equal numbers of active tracks and detections. Pipeline time is the sum of the measured tracker time and the fixed 34.8-ms detector cost; FPS is its reciprocal.

Tracks/detections	Tracker (ms)	Pipeline (ms)	Pipeline FPS
5	2.27	37.07	26.98
10	3.14	37.94	26.36
20	5.10	39.90	25.06
40	9.37	44.17	22.64
80	20.84	55.64	17.97